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Cloud Computing
BCA — Semester 7 — 2024 — Model Answer Sheet
Subject Code: CACS402

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define cloud computing. Explain the essential characteristics of cloud computing.

Answer: Cloud Computing: Delivery of computing services (servers, storage, databases, networking, software) over the Internet ('the cloud'). **Essential Characteristics (NIST):** 1) **On-demand self-service:** Users provision resources automatically without human interaction. 2) **Broad network access:** Available over network through standard mechanisms. 3) **Resource pooling:** Multi-tenant model with dynamic allocation. 4) **Rapid elasticity:** Resources scale up/down automatically. 5) **Measured service:** Pay-per-use metering.

3. Differentiate between IaaS, PaaS, and SaaS with examples.

Answer: IaaS (Infrastructure as a Service): Provides virtualized computing resources. User manages OS, applications, data. Examples: AWS EC2, Azure VMs, DigitalOcean. **PaaS (Platform as a Service):** Provides development platform with tools, libraries, middleware. User manages applications and data. Examples: Heroku, Google App Engine, AWS Elastic Beanstalk. **SaaS (Software as a Service):** Complete application delivered over internet. Provider manages everything. Examples: Gmail, Salesforce, Microsoft 365. **Comparison:** IaaS offers most control; SaaS offers least control but easiest to use. PaaS balances control and convenience for developers.

4. Explain the cloud deployment models: public, private, hybrid, and community cloud.

Answer: Public Cloud: Owned by cloud provider, shared among multiple tenants. Cost-effective, scalable, no maintenance. Examples: AWS, Azure, GCP. **Private Cloud:** Dedicated to single organization. Greater control, security, compliance. Higher cost. Can be on-premises or hosted. Examples: VMware, OpenStack. **Hybrid Cloud:** Combines public and private. Sensitive data on private, scalable workloads on public. Examples: AWS Outposts, Azure Stack. **Community Cloud:** Shared among organizations with common concerns. Examples: Government cloud, healthcare cloud. **Selection Criteria:** Security needs, compliance, budget, scalability requirements.

5. What is virtualization? Explain the types of virtualization used in cloud computing.

Answer: Virtualization: Creating virtual versions of physical resources (servers, storage, networks). **Types:** **Server Virtualization:** Multiple OS on single physical server using hypervisors (VMware, Hyper-V). **Storage Virtualization:** Pooling physical storage into virtual volumes (SAN, NAS). **Network Virtualization:** Abstracting network resources (SDN, VLANs). **Desktop Virtualization:** Hosting desktops centrally (VDI).

Application Virtualization: Running apps in isolated environments (Docker containers). **Hypervisors:** Type 1 (bare metal: VMware ESXi) and Type 2 (hosted: VirtualBox). **Benefits:** Resource efficiency, isolation, portability, disaster recovery.

6. Describe the architecture of cloud computing. Explain the role of frontend and backend.

Answer: Cloud Architecture Components: **Frontend (Client Side):** User interface and client infrastructure. Includes web browsers, mobile apps, thin/thick clients. Interacts with cloud through APIs. **Backend (Cloud Side):** All resources providing services. Includes servers, storage, databases, security mechanisms, traffic control, middleware. **Key Components:** **Hypervisor:** Manages virtual machines. **Management Software:** Allocates resources, monitors performance. **Deployment Software:** Deploys applications. **Network:** Connects components. **Storage:** Persistent data storage. **Security:** Firewalls, encryption, identity management.

7. What is load balancing? Explain different load balancing algorithms used in cloud computing.

Answer: Load Balancing: Distributing incoming network traffic across multiple servers to prevent any single server from becoming overloaded. **Algorithms:** **Round Robin:** Requests distributed sequentially. **Least Connections:** Sent to server with fewest active connections. **Least Response Time:** Sent to fastest-responding server. **IP Hash:** Client IP determines server (session persistence). **Weighted Methods:** Servers with more capacity get more traffic. **Benefits:** High availability, fault tolerance, scalability, optimal resource utilization. **Implementation:** Hardware load balancers (F5), software (Nginx, HAProxy), cloud-native (AWS ELB, Azure Load Balancer).

8. Explain the security challenges in cloud computing and how to address them.

Answer: Security Challenges: **Data Breaches:** Unauthorized access to sensitive data. **Insider Threats:** Malicious employees of cloud provider. **Account Hijacking:** Credential theft. **Insecure APIs:** Poorly designed interfaces. **Denial of Service:** Overwhelming resources. **Compliance:** Meeting regulatory requirements. **Solutions:** Encryption at rest and in transit. Strong IAM with MFA. Regular security audits and penetration testing. Data backup and disaster recovery. Compliance certifications (SOC 2, ISO 27001). Shared responsibility model (provider secures infrastructure, customer secures data/applications).

Group C

Attempt any TWO questions. [2×10=20]

9. Compare AWS, Azure, and Google Cloud Platform. Discuss their services, pricing models, and use cases. Which platform would you recommend for a startup and why?

Answer: AWS: Most mature, largest service catalog (200+ services). Extensive global infrastructure. Pay-as-you-go pricing. Best for enterprises, complex workloads. **Azure:** Strong Microsoft integration (Office 365, Windows Server). Hybrid cloud capabilities. Good for enterprises using Microsoft stack. **GCP:** Best-in-class data analytics and AI/ML (BigQuery, TensorFlow). Kubernetes leadership. Competitive pricing. Best for data-intensive and AI applications. **Pricing:** All use pay-as-you-go with reserved instance discounts. GCP often has lowest compute prices. **Startup Recommendation:** AWS for broadest ecosystem and documentation; GCP for data/AI focus and startup credits; Azure if using Microsoft tools. Consider multi-cloud strategy to avoid vendor lock-in.

10. Explain the concept of serverless computing. How does AWS Lambda work? Write a simple Lambda function and explain its deployment and execution.

Answer: Serverless Computing: Cloud provider manages server provisioning and allocation. Developers write functions triggered by events. Pay only for execution time. **AWS Lambda:** Event-driven serverless platform. **How it works:** 1) Upload function code (Python, Node.js, Java, etc.). 2) Configure trigger (API Gateway, S3

event, CloudWatch schedule). 3) Lambda creates execution environment when triggered. 4) Function runs, returns result. 5) Environment may be reused for subsequent calls (warm start). **Example (Python):** `def lambda_handler(event, context): return {'statusCode': 200, 'body': 'Hello from Lambda!'}`. **Deployment:** Via AWS Console, CLI, or SAM/CDK. **Limitations:** 15-minute timeout, 10GB memory, cold start latency.

11. What is Docker and Kubernetes? Explain containerization and orchestration. How do they improve application deployment in cloud environments?

Answer: Docker: Platform for developing, shipping, and running applications in containers. **Containerization:** Packages application with dependencies into isolated, portable units. Containers share OS kernel but have isolated filesystem, network, processes. Lighter than VMs. **Kubernetes:** Open-source container orchestration platform. Automates deployment, scaling, and management. **Key Concepts:** Pods (smallest deployable units), Services (network access), Deployments (declarative updates), ReplicaSets (maintain desired pod count). **Benefits:** Consistent environment across dev/test/prod. Rapid scaling (horizontal pod autoscaling). Self-healing (restarts failed containers). Rolling updates with zero downtime. Resource efficiency. Cloud-native applications.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.